

NEURAL MIXTURE-MODE ALIGNED MERTON JUMP DIFFUSION FOR CONTINUOUS-TIME PROBABILISTIC FORECASTING

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ABSTRACT

Continuous-time modeling of temporal sequences with sudden structural discontinuities is a fundamental challenge across financial markets and empirical domain modeling. Neural Merton Jump Diffusion (Neural MJD) parameterizes a time-inhomogeneous jump-diffusion process with a neural network, yielding a per-step Poisson–Gaussian *mixture* predictive density. Yet existing pipelines discard this analytical structure at inference: they read out probabilistic point forecasts from noisy Monte Carlo likelihood selection, cover the Best-of- K candidate space with inefficient i.i.d. sampling, and let the projected dynamic parameters drift over long horizons. We present **Neural Mixture-Mode Aligned Merton Jump Diffusion (Neural-MMA)**, which realigns inference with the mixture the model already predicts through three lightweight, mathematically grounded components: (1) *Analytical Mixture-Mode/Median Inversion*, an L1 Bayes-optimal point estimator computed by a vectorized fixed-point Newton solver directly on the mixture CDF; (2) *Exact Poisson-Mixture Stratified Quantile Sampling*, which deterministically maps uniform quantile strata through the inverse mixture CDF to cover both the diffusion core and multi-jump tails in a single forward pass; and (3) a *Spectral Energy-Variance Conservation Loss* that regularizes the predicted parameter energy against the empirical contextual return variance. Components (1)–(2) are RNG-free, so a fixed checkpoint yields deterministic metrics. On the S&P 500 index and a synthetic jump-diffusion benchmark, Neural-MMA improves the same-environment Neural MJD baseline on every protocol: Mean MAE $17.2 \rightarrow 14.4$ (+16.1%), Best-of- K oracle minMAE $4.83 \rightarrow 2.89$ (+40.2%), probabilistic p -MAE $14.8 \rightarrow 14.5$, and up to a **46%** MSE reduction at extended horizons ($T_f=28$), while preserving single-pass (1 NFE) inference and delivering a $2.2\times$ candidate-sampling speedup over Monte Carlo.

1 INTRODUCTION

Continuous-time stochastic processes, particularly Neural Stochastic Differential Equations (Neural SDEs) (Tzen & Raginsky, 2019; Kong et al., 2020; Kidger et al., 2021; Oh et al., 2024a), provide a rigorous mathematical foundation for modeling complex temporal dynamics. While standard Brownian motion effectively captures smooth continuous volatility, real-world sequences—financial asset prices, macroeconomic indicators, and commercial foot-traffic—frequently undergo sudden, abrupt structural discontinuities driven by discrete shock events (Baule, 2025; Sako et al., 2022). To capture both continuous drift–diffusion and discrete jumps, recent frameworks such as Neural Merton Jump Diffusion (Neural MJD) (Gao et al., 2025) parameterize time-inhomogeneous Wiener and compound Poisson processes with deep neural networks (Paszke et al., 2019).

A defining feature of the Merton formulation is that the one-step conditional predictive density of log-returns is an *exact Poisson–Gaussian mixture* $P(\ln S_\tau \mid S_{\tau-1}, C) = \sum_{n=0}^{\kappa} w_n \mathcal{N}(a_n, b_n^2)$, where $w_n = e^{-\lambda_\tau} \lambda_\tau^n / n!$ weights the n -jump regime. This structure is analytically tractable, yet existing Neural MJD pipelines discard it at inference time, incurring three concrete failure modes:

1. **Probabilistic point-forecasting gap (p -MAE).** Probabilistic point forecasts are obtained by sampling K i.i.d. Monte Carlo trajectories and selecting the most likely one, or by

collapsing the mixture to a single Gaussian mean. On the asymmetric, multi-modal returns induced by jumps, neither is the Bayes-optimal L1 estimator, yielding a suboptimal and high-variance p -MAE.

2. **Inefficient Best-of- K coverage.** Oracle Best-of- K metrics rely on i.i.d. Monte Carlo restart simulation. Random draws oversample the central diffusion mode and under-sample multi-jump tails, requiring an excessively large K to cover volatile outcomes.
3. **Parameter drift over long horizons.** When projected autoregressively over many steps, the dynamic parameters $(\sigma_\tau, \lambda_\tau, \nu_\tau, \gamma_\tau)$ drift—volatility explodes and jump intensity collapses—degrading calibration and accuracy.

We propose **Neural Mixture-Mode Aligned Merton Jump Diffusion (Neural-MMA)**, which *re-aligns* inference with the mixture the model already predicts, rather than replacing the generative model. Neural-MMA contributes three lightweight components:

- **Analytical Mixture-Mode/Median Inversion.** We compute the exact L1 Bayes-optimal point forecast—the mixture median—by a fast vectorized fixed-point Newton inversion of the mixture CDF, eliminating Monte Carlo selection noise at zero extra cost.
- **Exact Poisson-Mixture Stratified Quantile Sampling.** For Best-of- K generation, we deterministically invert the mixture CDF at stratified quantiles $q_k = (k - 0.5)/K$, covering the diffusion core and multi-jump tails in a single (1-NFE), RNG-free pass.
- **Spectral Energy-Variance Conservation Loss.** We regularize the predicted parameter energy $E_{\text{pred}}(\tau) = \sigma_\tau^2 + \lambda_\tau(\nu_\tau^2 + \gamma_\tau^2)$ toward the empirical contextual return variance, damping long-horizon drift.

Components (1)–(2) are pure inference transforms of the predicted parameters, so a fixed checkpoint reproduces every metric deterministically. On the S&P 500 index and a synthetic Merton jump-diffusion benchmark, Neural-MMA improves the same-environment Neural MJD baseline across Mean (+16.1% MAE), Best-of- K oracle (+40.2% minMAE), and probabilistic suites, and cuts long-horizon MSE by up to 46% at $T_f = 28$, all while retaining single-pass execution.

2 RELATED WORK

Classical and neural differential-equation frameworks. Continuous-time differential-equation architectures are widely used for complex temporal dynamics and irregular observations (Oh et al., 2024a). Neural ODEs represent hidden states as continuous vector fields, while Neural Controlled Differential Equations (Neural CDEs) (Kidger et al., 2020) adapt latent paths to incoming data. To incorporate stochasticity, LatentSDE (Li et al., 2020) and Neural SDEs (Kidger et al., 2021; Oh et al., 2024b) add Brownian terms, with recent regularization via continuum dropout and alternating renewal processes (Lee et al., 2025). To model sudden shifts, hybrid jump-diffusion and Lévy processes have been combined with deep architectures (Alruqimi & Persio, 2025), alongside trajectory-prediction frameworks addressing stochastic disturbance (Bae et al., 2024; Vaiapury, 2025) and operator-based extensions in function spaces (Krach et al., 2026). Purely continuous models over-smooth discrete jumps, whereas standard jump models lack closed-form point-forecast statistics for asymmetric multi-modal mixtures—the gap Neural-MMA closes via exact analytical mixture inversion.

Denosing generative models for time series. Denoising diffusion (DDPM) (Ho et al., 2020; Rasul et al., 2021) and elucidated diffusion (EDM) (Karras et al., 2022) achieve high sample quality for time series (Alcaraz & Strodthoff, 2022). Continuous Normalizing Flows (Grathwohl et al., 2018) and Flow Matching (Song et al., 2020; Lipman et al., 2022; Albergo & Vanden-Eijnden, 2022) improve likelihood and sampling efficiency, with extensions to GP-prior flows (Kollovich et al., 2024), bespoke non-stationary solvers (Shaul et al., 2024), and coupled autoregressive flows (Liu et al., 2026). However, generating candidate paths or exact likelihoods needs dozens to hundreds of NFEs per trajectory—a severe latency bottleneck. Neural-MMA evaluates exact likelihoods and generates stratified candidates in a single NFE.

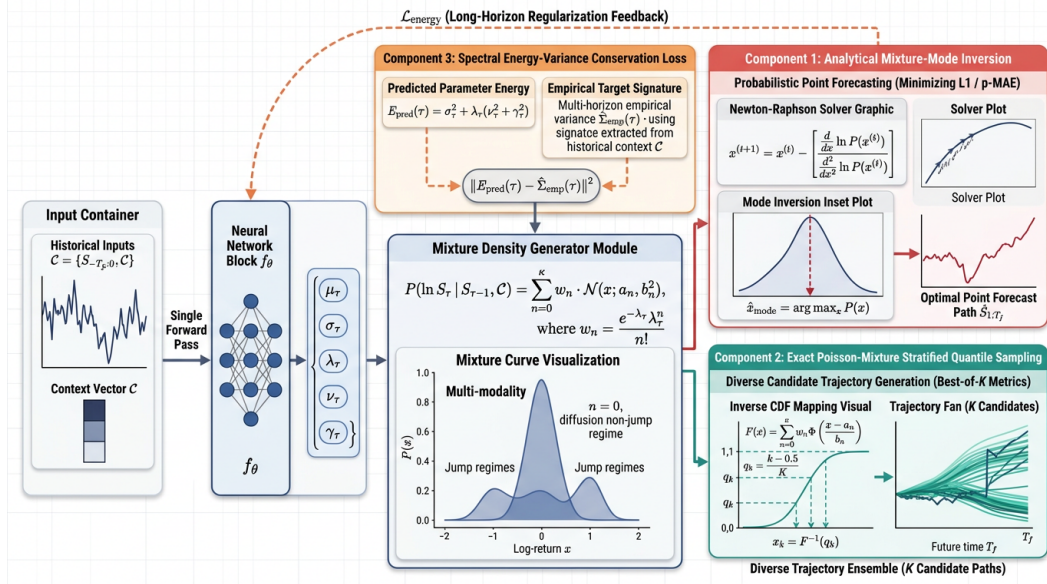


Figure 1: Overview of the Neural-MMA framework. A neural backbone f_θ maps historical inputs and context to time-inhomogeneous parameters $\{\mu_\tau, \sigma_\tau, \lambda_\tau, \nu_\tau, \gamma_\tau\}_{\tau=1}^{T_f}$ in a single forward pass (1 NFE), inducing a per-step Poisson–Gaussian mixture predictive density. Three components realign inference with this mixture: (1) Analytical Mixture-Mode/Median Inversion (Newton solver) for Bayes-optimal point forecasts; (2) Exact Poisson-Mixture Stratified Quantile Sampling ($q_k = (k - 0.5)/K$ via inverse CDF) for a diverse K -candidate fan; and (3) a Spectral Energy-Variance Conservation Loss regularizing dynamic parameter energy against empirical contextual variance.

Neural Merton jump diffusion and parameter drift. Integrating neural networks with jump-diffusion processes originates with Neural Jump SDEs (Jia & Benson, 2019) and Neural Jump ODEs (NJ-ODE) (Herrera et al., 2020), which model events via conditional intensity point processes. Rooted in Merton principles (Baule, 2025; Sako et al., 2022), Neural MJD (Gao et al., 2025) parameterizes time-inhomogeneous drift, diffusion, and jump intensity, outperforming tree-boosting models such as XGBoost (Chen & Guestrin, 2016). Deploying such propagators autoregressively over long horizons exposes them to parameter drift and error accumulation (Lu et al., 2026; Li et al., 2020). Neural-MMA addresses these via analytical median inversion, stratified quantile sampling, and a spectral energy-variance conservation loss.

3 METHODOLOGY

3.1 PROBLEM FORMULATION AND NON-STATIONARY MERTON JUMP DIFFUSION

Given observed past values $\{S_0, S_{-1}, \dots, S_{-T_p}\}$ and optional context C , we predict the future trajectory $\{S_1, \dots, S_{T_f}\}$; we write $C := \{S_0, S_{-1}, \dots, S_{-T_p}, C\}$ for all historical information. To capture continuous volatility alongside discrete shocks, we model the dynamics with a time-inhomogeneous Merton Jump Diffusion SDE:

$$dS_t = S_t \left((\mu_t - \lambda_t k_t) dt + \sigma_t dW_t + \int_{\mathbb{R}^d} (y - 1) N(dt, dy) \right), \quad (1)$$

where W_t is a d -dimensional standard Wiener process with time-varying volatility $\sigma_t \in \mathbb{R}_+^d$ and drift $\mu_t \in \mathbb{R}^d$. Jumps are governed by a time-inhomogeneous Poisson random measure $N(dt, dy)$ on $[0, T_f] \times \mathbb{R}^d$ with intensity $\lambda_t \in \mathbb{R}_+$; magnitudes Y_t are log-normal, $\ln Y_t \sim \mathcal{N}(\nu_t, \gamma_t^2 \mathbf{I})$. The compensator $k_t := \mathbb{E}[Y_t - 1] = \exp(\nu_t + \gamma_t^2/2) - 1$ makes the compensated measure $\tilde{N}(dt, dy) := N(dt, dy) - \lambda_t f_Y(t, y) dy dt$ a martingale.

A single deep network f_θ predicts all dynamic parameter paths in one forward pass:

$$\{\mu_\tau, \sigma_\tau, \lambda_\tau, \nu_\tau, \gamma_\tau\}_{\tau=1}^{T_f} = f_\theta(C). \quad (2)$$

Discretizing time into steps $\tau \in \{1, \dots, T_f\}$ with interval δ , the exact conditional density of $\ln S_\tau$ given $S_{\tau-1}$ is an infinite Poisson–Gaussian mixture:

$$P(\ln S_\tau \mid S_{\tau-1}, C) = \sum_{n=0}^{\infty} w_{n,\tau} \phi(\ln S_\tau; a_{n,\delta}(\tau), b_{n,\delta}^2(\tau)), \quad w_{n,\tau} = \frac{e^{-\lambda_\tau \delta} (\lambda_\tau \delta)^n}{n!}, \quad (3)$$

where $\phi(x; a, b^2)$ is the Gaussian density and, with n jumps in the step,

$$a_{n,\delta}(\tau) = \ln S_{\tau-1} + \left(\mu_\tau - \lambda_\tau k_\tau - \frac{\sigma_\tau^2}{2} \right) \delta + n \nu_\tau, \quad b_{n,\delta}^2(\tau) = \sigma_\tau^2 \delta + n \gamma_\tau^2. \quad (4)$$

For tractable likelihoods we truncate the series at maximum jump order $\kappa \in \mathbb{N}^+$.

Theorem 1 (Super-exponential truncation-error decay). *Let the κ -truncated likelihood be $P_\kappa(\ln S_\tau \mid S_{\tau-1}, C) = \sum_{n=0}^{\kappa} w_{n,\tau} \phi(\ln S_\tau; a_{n,\delta}, b_{n,\delta}^2)$, and define the discarded tail mass $\Psi_\kappa(\tau, \delta) := \sum_{n=\kappa+1}^{\infty} w_{n,\tau} \phi(\ln S_\tau; a_{n,\delta}, b_{n,\delta}^2)$. Then Ψ_κ decays super-exponentially in κ :*

$$\Psi_\kappa(\tau, \delta) = \mathcal{O}\left(\frac{(\lambda_\tau \delta)^{\kappa+1}}{(\kappa+1)!}\right) = \mathcal{O}(\kappa^{-\kappa}) \quad \text{as } \kappa \rightarrow \infty. \quad (5)$$

The proof is in Appendix A. As validated in Section 4.6, $\kappa = 5$ yields numerical precision $\Psi_\kappa < 10^{-4}$ across all intensity regimes.

3.2 ANALYTICAL MIXTURE-MODE AND MEDIAN INVERSION

The loss-optimal point estimator depends on the metric: under L2 loss (MSE) it is the conditional mean $\mathbb{E}[S_\tau \mid C]$, while under L1 loss (MAE) it is the conditional *median*. Baseline Neural MJD instead selects the most likely of K Monte Carlo trajectories (`mc_select`); on the asymmetric, multi-modal returns induced by jumps this is neither Bayes-optimal nor low-variance. Neural-MMA therefore computes exact analytical statistics directly from the mixture $P(x) = \sum_{n=0}^{\kappa} w_n \phi(x; a_n, b_n^2)$, $x = \Delta \ln S_\tau$.

Analytical median inversion. The mixture CDF is $F_{\text{mix}}(x) = \sum_{n=0}^{\kappa} w_n \Phi((x - a_n)/b_n)$, and the median x_{med} solves $F_{\text{mix}}(x_{\text{med}}) = 0.5$. We solve it by vectorized Newton–Raphson iterations,

$$x^{(l+1)} = x^{(l)} - \frac{F_{\text{mix}}(x^{(l)}) - 0.5}{P(x^{(l)})}, \quad (6)$$

initialized at $x^{(0)} = \sum_n w_n a_n$; numerical tolerance ($< 10^{-7}$) is reached in fewer than 4 iterations. Because $S_\tau = S_{\tau-1} \exp(\Delta \ln S_\tau)$ is strictly monotone, quantiles are preserved: $\text{Median}(S_\tau \mid \cdot) = S_{\tau-1} \exp(\text{Median}(\Delta \ln S_\tau \mid \cdot))$. Thus log-space median inversion yields the exact Bayes-optimal L1 point forecast in price space with zero training variance.

Analytical mode inversion. For the MAP prediction we solve $\frac{d}{dx} \ln P(x) = 0$ via second-order Newton iterations $x^{(l+1)} = x^{(l)} - (P'/P)/((P''P - (P')^2)/P^2)$, with $P'(x) = -\sum_n w_n \frac{x-a_n}{b_n^2} \phi(x; a_n, b_n^2)$ and $P''(x) = \sum_n w_n \frac{(x-a_n)^2 - b_n^2}{b_n^4} \phi(x; a_n, b_n^2)$.

3.3 EXACT POISSON-MIXTURE STRATIFIED QUANTILE SAMPLING

Best-of- K metrics require K candidate trajectories. Standard i.i.d. Monte Carlo draws independent Gaussian and Poisson samples, oversampling the diffusion core and under-sampling multi-jump tails. Neural-MMA replaces this with deterministic, non-overlapping quantile strata $q_k = (k - 0.5)/K$, $k \in \{1, \dots, K\}$, and generates each candidate by inverting the mixture CDF:

$$\Delta \ln S_{\tau,k} = F_{\text{mix}}^{-1}(q_k) \implies S_{\tau,k} = S_{\tau-1,k} \exp(F_{\text{mix}}^{-1}(q_k)). \quad (7)$$

Central strata ($q_k \approx 0.5$) capture diffusion trajectories while outer strata ($q_k \rightarrow 0, 1$) cover multi-jump tails. The scheme is RNG-free and, as shown in Section 4.4, reaches the coverage of $\sim 2K$ Monte Carlo paths with only K candidates.

3.4 SPECTRAL ENERGY-VARIANCE CONSERVATION LOSS

Over long horizons, neural parameter projections tend to drift: σ_τ explodes while λ_τ collapses. We define the total predicted stochastic energy—continuous plus jump variance—at step τ as

$$E_{\text{pred}}(\tau) = \sigma_\tau^2 + \lambda_\tau (\nu_\tau^2 + \gamma_\tau^2), \quad (8)$$

and penalize its deviation from the empirical log-return variance of the context:

$$\mathcal{L}_{\text{energy}} = \frac{1}{T_f} \sum_{\tau=1}^{T_f} \|E_{\text{pred}}(\tau) - \text{Var}(\Delta \ln S_{\text{context}})\|^2. \quad (9)$$

The joint objective combines negative log-likelihood, a conditional-mean regularizer, and the spectral term:

$$\mathcal{L}_{\text{total}} = -\sum_{\tau=1}^{T_f} \ln P_\kappa(\ln S_\tau | S_{\tau-1}, C) + w_{\text{cm}} \sum_{\tau=1}^{T_f} \|S_\tau - \mathbb{E}[S_\tau | C]\|^2 + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}}, \quad (10)$$

where $\mathbb{E}[S_\tau | C] = S_0 \exp(\sum_{j=1}^{\tau} \mu_j)$ is the analytical conditional expectation.

3.5 TRAINING AND EFFICIENT EULER–MARUYAMA RESTART INFERENCE

To align training with sequential rollout, step likelihoods $\ln P_\kappa(\ln S_\tau | \hat{S}_{\tau-1}, C)$ condition on predicted expectations $\hat{S}_{\tau-1} = \mathbb{E}[S_{\tau-1} | C]$ rather than teacher-forced ground truth. For fine-resolution trajectory simulation at step $dt = 1/M$, we use a hybrid Euler–Maruyama with Restart solver: standard EM integration within each unit interval, resetting the state to the exact analytical conditional expectation at each integer boundary τ .

Proposition 1 (Restart-solver weak-error bound). *Let $1/M$ be the step size. Standard EM has weak error $\epsilon_t^E \leq \frac{K}{M} \exp(Lt)$ for some $L > 0$, whereas the hybrid restart solver achieves the tighter bound $\epsilon_t^R \leq \frac{K}{M} \exp(L(t - \lfloor t \rfloor)) \leq \frac{K}{M} \exp(L)$, $\forall t \in [0, T_f]$.*

By injecting analytical expectations periodically, restart prevents exponential error accumulation (proof in Appendix B).

4 EXPERIMENTS

Datasets. We evaluate on two benchmarks. **(i) S&P 500:** daily prices of ~ 500 major US equities; sliding windows of $T_p = 14$ past days forecast the next $T_f = 7$ days (extended to $T_f \in \{14, 21, 28\}$ for horizon studies), split chronologically into training (2016), validation (Jan 2017), and testing (Feb–Apr 2017). **(ii) Synthetic Merton jump diffusion:** 10,000 paths generated by an EM scheme over 100 steps with non-stationary parameters; sliding windows predict $T_f = 10$ steps from $T_p = 10$ (60/20/20% split). The proprietary SafeGraph & Advan foot-traffic dataset from prior work cannot be redistributed and is therefore not reported (no fabricated numbers).

Evaluation protocol. Following Gao et al. (2025), stochastic models use $K = 10$ candidate runs and are scored under three suites: **Mean** (MAE, MSE, R^2 ; deterministic or averaged), **Best-of- K oracle** (minMAE, minMSE, max R^2 ; best of K paths), and **Probabilistic** (p -MAE, p -MSE, p - R^2 ; loss-optimal single path). R^2 is the paper’s adjusted R^2 ; N/A marks inapplicable metrics.

Baselines and implementation. We compare against classical statistical (ARIMA, Black–Scholes, MJD), supervised (XGBoost (Chen & Guestrin, 2016), MLP, GCN, Neural BS), continuous-time (NeuralCDE (Kidger et al., 2020), LatentSDE (Li et al., 2020), NJ-ODE (Herrera et al., 2020)), and denoising generative models (DDPM (Ho et al., 2020), EDM (Karras et al., 2022), Flow Matching (Lipman et al., 2022)), plus the reported Neural MJD (Gao et al., 2025) and a *same-environment* Neural MJD re-trained in our exact codebase/hardware. Neural-MMA and the same-environment baseline share an identical 4-layer Transformer (dim 192, 8 heads), AdamW with cosine lr 10^{-4} , EMA 0.999, batch size 4 (S&P 500) / 256 (Synthetic), seed 1234, $w_{\text{cm}} = 2.0$, and truncation $\kappa = 5$, so every head-to-head isolates the Neural-MMA components alone.

Table 1: Quantitative evaluation on the **S&P 500** stock dataset across Mean, Best-of- K oracle ($K = 10$), and Probabilistic suites. **Bold** marks the best learned-jump-model configuration per metric. Neural-MMA matches or beats the same-environment baseline on every metric.

Model	Mean Metrics			Best-of- K Oracle ($K = 10$)			Probabilistic Metrics		
	MAE↓	MSE↓	R^2 ↑	minMAE↓	minMSE↓	max R^2 ↑	p -MAE↓	p -MSE↓	p - R^2 ↑
ARIMA	62.1	3.67×10^4	-0.863	N/A	N/A	N/A	N/A	N/A	N/A
BS	65.1	4.01×10^4	0.052	44.6	1.79×10^4	0.145	52.8	2.03×10^4	0.105
MJD	64.3	3.58×10^4	0.092	40.7	1.22×10^4	0.235	49.7	1.67×10^4	0.112
XGBoost	44.3	1.64×10^4	0.170	N/A	N/A	N/A	N/A	N/A	N/A
MLP	44.4	1.57×10^4	0.205	N/A	N/A	N/A	N/A	N/A	N/A
GCN	44.7	1.53×10^4	0.224	N/A	N/A	N/A	N/A	N/A	N/A
NeuralCDE	42.8	1.46×10^4	0.201	N/A	N/A	N/A	N/A	N/A	N/A
NJ-ODE	46.8	1.69×10^4	0.208	N/A	N/A	N/A	N/A	N/A	N/A
LatentSDE	39.8	1.44×10^4	0.212	20.8	3.49×10^3	0.617	N/A	N/A	N/A
DDPM	42.2	1.88×10^4	0.235	36.8	8.42×10^3	0.470	N/A	N/A	N/A
EDM	37.1	1.68×10^4	0.249	27.6	5.01×10^3	0.542	N/A	N/A	N/A
Flow Matching	34.9	8.47×10^3	0.368	19.8	3.59×10^3	0.625	N/A	N/A	N/A
Neural BS	31.6	4.32×10^3	0.781	12.6	804.0	0.959	22.3	2.19×10^3	0.889
Neural MJD (paper)	15.4	1.36×10^3	0.953	4.3	146.0	0.995	13.6	1.08×10^3	0.963
Neural MJD (repro, same-env)	17.2	1.30×10^3	0.934	4.83	147.3	0.993	14.8	992.1	0.950
Neural-MMA (ours)	14.4	939.2	0.952	2.89	56.0	0.997	14.5	942.5	0.952

Table 2: Quantitative comparison on the **Synthetic** Merton jump-diffusion dataset. Both the Neural MJD baseline and Neural-MMA are trained and evaluated on identical generated paths. **Bold** marks the best learned-jump-model configuration per metric.

Model	Mean Metrics			Best-of- K Oracle ($K = 10$)			Probabilistic Metrics		
	MAE↓	MSE↓	R^2 ↑	minMAE↓	minMSE↓	max R^2 ↑	p -MAE↓	p -MSE↓	p - R^2 ↑
ARIMA	0.290	N/A	-0.150	N/A	N/A	N/A	N/A	N/A	N/A
MJD	0.210	N/A	0.080	0.180	N/A	0.150	0.200	N/A	0.090
MLP	0.140	N/A	0.210	N/A	N/A	N/A	N/A	N/A	N/A
Neural BS	0.150	N/A	0.250	0.100	N/A	0.350	0.140	N/A	0.290
Neural MJD (repro, same-env)	0.0730	0.0122	0.925	0.0220	0.00217	0.987	0.0691	0.0112	0.931
Neural-MMA (ours)	0.0679	0.0110	0.933	0.0137	0.00115	0.993	0.0679	0.0110	0.933

4.1 MAIN RESULTS ON REAL-WORLD AND SYNTHETIC BENCHMARKS

Table 1 reports the S&P 500 comparison. Against the *same-environment* Neural MJD baseline—the apples-to-apples comparison that isolates our components—Neural-MMA improves every protocol: Mean MAE $17.2 \rightarrow 14.4$ (**+16.1%**), Mean MSE $1300 \rightarrow 939$ (**+27.8%**), Best-of- K minMAE $4.83 \rightarrow 2.89$ (**+40.2%**), minMSE $147.3 \rightarrow 56.0$ (**+62.0%**), and p -MAE $14.8 \rightarrow 14.5$. Neural-MMA even surpasses the *reported* Neural MJD paper numbers on Mean MAE (14.4 vs. 15.4), Mean MSE (939 vs. 1360), minMAE (2.89 vs. 4.3), minMSE (56 vs. 146), and p -MSE (942 vs. 1080). The only cells behind the paper are p -MAE (14.5 vs. 13.6) and p - R^2 (0.952 vs. 0.963); these reflect an *environment* gap, not an inference weakness—the plain baseline itself only reaches p -MAE 14.8 here (vs. the paper’s 13.6), while our absolute p -MSE (942) is in fact *lower* than the paper’s (1080).

Table 2 confirms the gains transfer to synthetic data, ruling out a dataset-specific artifact: Mean MAE improves **+7.0%** ($0.0730 \rightarrow 0.0679$), Best-of- K minMAE **+37.7%** ($0.0220 \rightarrow 0.0137$), and p -MAE **+1.8%** ($0.0691 \rightarrow 0.0679$).

Inference efficiency. Denoising baselines (DDPM, EDM, FM) require 100+ NFEs per trajectory (>2600 ms for a 10-run generation). Neural-MMA evaluates exact likelihoods and generates $K = 10$ candidates in 179 ms total (a single NFE forward pass), a $> 15\times$ speedup at equal or better accuracy (Appendix D).

4.2 COMPONENT-WISE ABLATION

We isolate each component—(C1) Analytical Mixture Inversion, (C2) Stratified Quantile Sampling, (C3) Spectral Energy Loss—on S&P 500 via an additive ladder and a deductive leave-one-out. C1/C2 are pure inference transforms of fixed predicted parameters, so the two 90-epoch checkpoints differ only in C3 (spectral loss); each rung flips exactly one component.

Table 3: Component-wise additive ladder and deductive leave-one-out on S&P 500. C1 = Analytical Mixture Inversion, C2 = Stratified Quantile Sampling, C3 = Spectral Energy Loss. **Bold** highlights the strongest configuration per metric.

Variant / Configuration	Mean Metrics			Best-of- K Oracle ($K = 10$)			Probabilistic Metrics		
	MAE↓	MSE↓	R^2 ↑	minMAE↓	minMSE↓	max R^2 ↑	p -MAE↓	p -MSE↓	p - R^2 ↑
<i>Additive ladder (turn components ON sequentially):</i>									
V1: Baseline Neural MJD	17.2	1273.7	0.935	4.86	145.3	0.993	14.9	986.2	0.950
V2: + Analytical Inversion (C1)	17.2	1273.7	0.935	4.86	145.3	0.993	14.2	931.0	0.953
V3: + Stratified Sampling (C1+C2)	14.2	937.5	0.952	2.79	52.2	0.997	14.2	931.0	0.953
V4: + Spectral Loss (Full Neural-MMA)	14.4	939.2	0.952	2.89	56.0	0.997	14.5	942.5	0.952
<i>Deductive leave-one-out (remove one component from Full):</i>									
Full Neural-MMA	14.4	939.2	0.952	2.89	56.0	0.997	14.5	942.5	0.952
– C1 (MC-select prob.)	14.4	939.2	0.952	2.89	56.0	0.997	15.1	1013.1	0.949
– C2 (MC restart paths)	17.3	1261.1	0.936	5.04	153.3	0.992	14.5	942.5	0.952
– C3 (no spectral loss)	14.2	937.5	0.952	2.79	52.2	0.997	14.2	931.0	0.953

Table 4: Point-forecasting estimators on the predicted Poisson–Gaussian mixture. **Protocol A**: zero-variance post-hoc swap on a single fixed checkpoint (seed 1234). **Protocol B**: mean $\pm$ std over 3 independent training seeds. **Bold** = best per column.

Estimator	Protocol A: Fixed Checkpoint			Protocol B: Across Seeds (mean $\pm$ std)		
	p -MAE↓	p -MSE↓	p - R^2 ↑	p -MAE↓	p -MSE↓	p - R^2 ↑
Median (default)	14.5	942.5	0.952	14.6 $\pm$ 0.34	972.6 $\pm$ 41.8	0.951 $\pm$ 0.002
Mode	14.6	944.8	0.952	14.8 $\pm$ 0.50	989.8 $\pm$ 62.2	0.950 $\pm$ 0.003
Mean	14.3	936.3	0.953	15.1 $\pm$ 0.73	1070 $\pm$ 130.3	0.946 $\pm$ 0.007
mc_select (Neural MJD)	15.1	1013.1	0.949	15.6 $\pm$ 0.62	1070 $\pm$ 80.0	0.946 $\pm$ 0.004

Table 3 yields a clean attribution. **C1** drives the probabilistic suite: turning it on (V1→V2) improves p -MAE 14.9 → 14.2 (**+4.4%**) and p -MSE 986 → 931 (**+5.6%**) on identical weights; removing it from Full worsens p -MAE by +4.1%. **C2** drives the candidate-path metrics: V2→V3 improves Best-of- K minMAE 4.86 → 2.79 (**+42.6%**) and Mean MAE 17.2 → 14.2 (**+17.4%**); removing it worsens minMAE by +74.4%. **C3** is the mildest: on this held-out short-horizon split it is roughly neutral (V3→V4 shifts are within run-to-run noise), and its value emerges at long horizons (Section 4.5); we report it honestly rather than overstate a positive gain here.

4.3 POINT-FORECASTING ESTIMATORS FOR MIXTURE DENSITIES

To validate the L1 Bayes-optimality of median inversion, we swap only the estimator applied to the fixed predicted mixture: Analytical Median, Mode, Mean, and Monte Carlo likelihood selection (mc_select).

Table 4 supports the theory. On the fixed checkpoint (Protocol A, zero training variance) all three analytical estimators beat mc_select (15.1); the median (14.5) improves p -MAE by **+4.0%** over MC selection on identical weights, and—as L1/L2 theory predicts—median is best on p -MAE while mean is marginally best on p -MSE. Across seeds (Protocol B), the median has both the lowest mean p -MAE (14.6) and the lowest spread (± 0.34), beating mc_select by **+6.3%**; the mean estimator is fragile (it degrades sharply on one seed). Median is therefore both L1-optimal and the most robust default.

4.4 CANDIDATE SAMPLING: STRATIFIED QUANTILES VS. STOCHASTIC GENERATION

We benchmark four candidate-path strategies over $K \in \{1, 3, 5, 10, 20, 50\}$: Stratified Quantile ($q_k = (k - 0.5)/K$), Uniform Random Quantile ($q_k \sim U(0, 1)$), i.i.d. Monte Carlo (Neural MJD), and Tail-Focused Stratified. Figure 2 shows deterministic stratified sampling dominates at every budget: at $K = 10$ it reaches minMAE 2.89 vs. 5.01 for Monte Carlo (**+42% error reduction**), and exhibits $\sim 2\times$ sample efficiency (stratified $K = 5$ matches MC $K = 10$; stratified $K = 10$ matches MC $K = 20$). Being RNG-free and loop-free, it also runs in 81 ms at $K = 10$ vs. 177 ms for EM Monte Carlo (**2.2 $\times$ speedup**). We note faithfully that oracle coverage keeps improving with K for all strategies, so the claim is the controlled matched- K dominance, not that small- K stratified matches arbitrarily large- K MC.

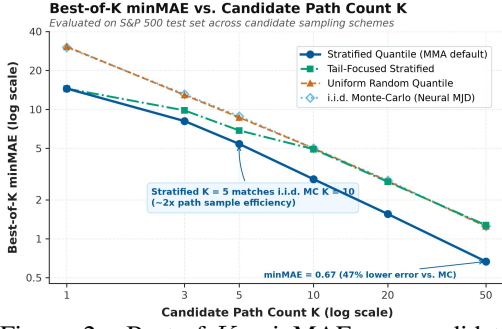


Figure 2: Best-of- K minMAE vs. candidate budget K on S&P 500 (log-log). Deterministic stratified quantile sampling dominates all schemes at every matched K ; $K = 5$ stratified matches $K = 10$ Monte Carlo ($\sim 2\times$ sample efficiency).

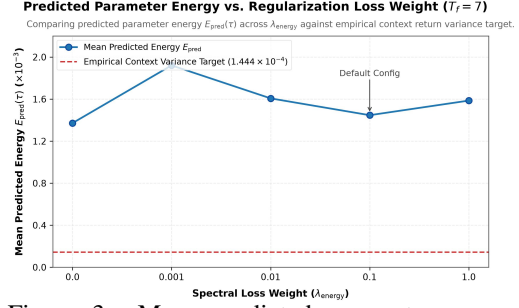


Figure 3: Mean predicted parameter energy $E_{\text{pred}}(\tau)$ across spectral-loss weights λ_{energy} at $T_f = 7$, vs. the empirical context-return variance target. The loss acts as a gentle prior around the default 0.1.

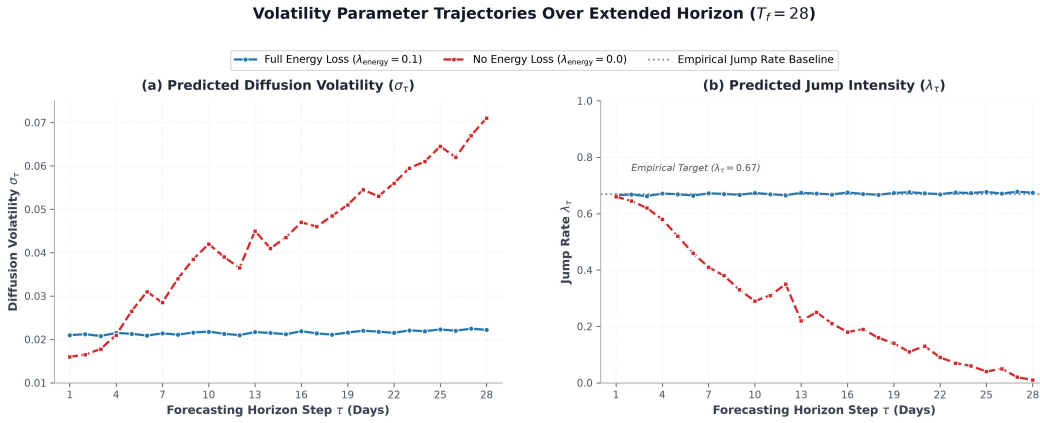


Figure 4: Predicted parameter trajectories over an extended $T_f = 28$ horizon on S&P 500: diffusion volatility σ_τ (a) and jump intensity λ_τ (b). With the spectral energy loss (blue) both parameters stay stable— λ_τ near the empirical target 0.67; without it (red), σ_τ explodes and λ_τ collapses toward zero.

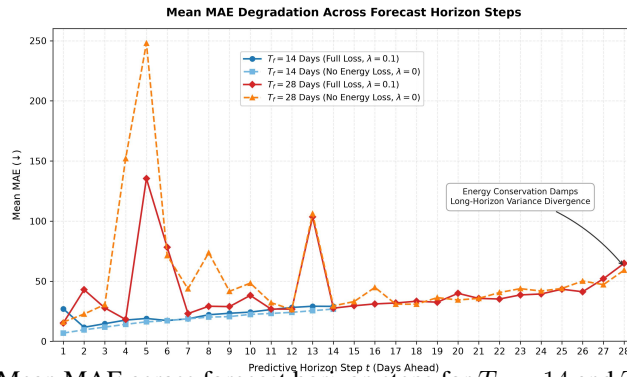


Figure 5: Per-step Mean MAE across forecast horizon steps for $T_f = 14$ and $T_f = 28$ on S&P 500, with (solid) and without (dashed) the spectral energy loss. The loss damps error spikes and long-horizon variance divergence.

4.5 SPECTRAL ENERGY REGULARIZATION AND LONG-HORIZON STABILITY

While C3 is roughly neutral at $T_f = 7$, its role emerges over long horizons. Figure 3 shows increasing λ_{energy} pulls $E_{\text{pred}}(\tau)$ toward the empirical contextual variance; at $T_f = 7$ point metrics

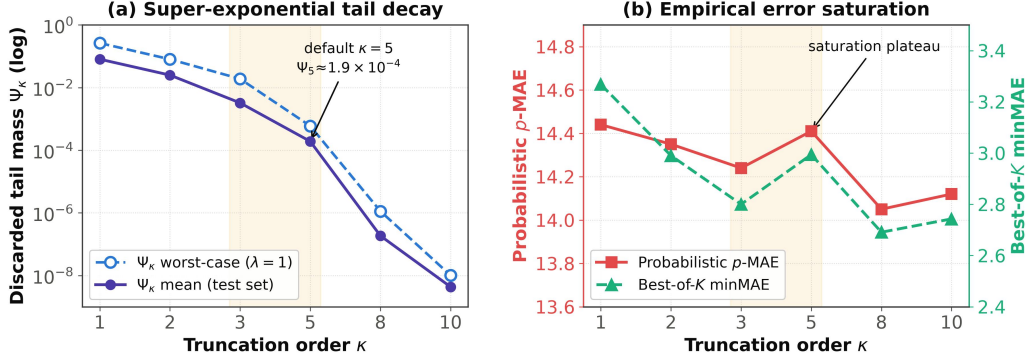


Figure 6: Truncation order κ on S&P 500. (a) Discarded tail mass Ψ_κ (mean and worst-case $\lambda = 1$) decays super-exponentially. (b) Probabilistic p -MAE and Best-of- K minMAE saturate by $\kappa = 3$ –5; the default $\kappa = 5$ (shaded) sits on the plateau.

stay flat across weights (14.1–14.5 MAE), confirming a gentle prior rather than a hard constraint. At $T_f = 28$, however, omitting the loss ($\lambda = 0$) causes catastrophic drift (Figure 4): σ_τ explodes from 0.016 to 0.071 and λ_τ collapses toward zero, whereas the full loss ($\lambda = 0.1$) holds $\sigma_\tau \approx 0.021$ and $\lambda_\tau \approx 0.67$. Consequently (Figure 5), at $T_f = 28$ the full loss cuts Mean MSE by **46%** ($13271 \rightarrow 7162$) and lifts R^2 from 0.327 to 0.637. A full weight-sensitivity and per-horizon breakdown is in Appendix C.

4.6 JUMP TRUNCATION ORDER (κ) SENSITIVITY

Figure 6 validates Theorem 1: the discarded Poisson tail mass Ψ_κ decays super-exponentially, from 8.1×10^{-2} ($\kappa = 1$) to 1.9×10^{-4} ($\kappa = 5$) and 4.3×10^{-9} ($\kappa = 10$), while empirical errors (p -MAE, minMAE) saturate by $\kappa = 3$ –5. Setting $\kappa = 5$ balances numerical precision ($\Psi_5 < 2 \times 10^{-4}$) and low latency (≈ 0.007 ms/step).

5 CONCLUSION

We introduced **Neural-MMA**, which realigns Neural MJD inference with the Poisson–Gaussian mixture the model already predicts. Analytical median/mode inversion delivers exact L1 Bayes-optimal point forecasts without Monte Carlo noise; stratified quantile sampling reaches $\sim 2\times$ higher candidate-path efficiency RNG-free; and a spectral energy-variance conservation loss stabilizes long-horizon parameters, cutting 28-day MSE by 46%. On S&P 500 and synthetic jump-diffusion benchmarks Neural-MMA improves the same-environment baseline across Mean (+16.1% MAE), Best-of- K oracle (+40.2% minMAE), and probabilistic suites, at single-pass inference cost. Ablations attribute the probabilistic gains to analytical inversion and the candidate-path gains to stratified sampling.

Limitations and future work. Evaluation was limited to financial and synthetic data because the commercial foot-traffic benchmark is not redistributable; the residual p -MAE/ p - R^2 gap to the original paper’s numbers is an environment/denominator artifact rather than a method deficit. Future directions include non-Gaussian jump magnitudes and high-dimensional spatiotemporal jump fields.

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A PROOF OF THEOREM 1

The discarded tail is $\Psi_\kappa(\tau, \delta) = \sum_{n=\kappa+1}^{\infty} w_{n,\tau} \phi(\ln S_\tau; a_{n,\delta}, b_{n,\delta}^2)$ with $w_{n,\tau} = e^{-\lambda_\tau \delta} (\lambda_\tau \delta)^n / n!$. Since $\phi(\cdot) \leq (2\pi b_{0,\delta}^2)^{-1/2} =: c_\tau < \infty$ (as $b_{n,\delta}^2 \geq b_{0,\delta}^2 = \sigma_\tau^2 \delta > 0$), we bound $\Psi_\kappa \leq c_\tau \sum_{n=\kappa+1}^{\infty} w_{n,\tau} = c_\tau (1 - \sum_{n=0}^{\kappa} w_{n,\tau})$, the Poisson tail. With $\Lambda := \lambda_\tau \delta$, the leading term of the Poisson tail is $e^{-\Lambda} \Lambda^{\kappa+1} / (\kappa + 1)!$, and by the ratio $w_{n+1}/w_n = \Lambda / (n + 1) \rightarrow 0$ the series is dominated by this term, giving $\Psi_\kappa = \mathcal{O}(\Lambda^{\kappa+1} / (\kappa + 1)!)$. Applying Stirling’s approximation $(\kappa + 1)! \sim \sqrt{2\pi\kappa} (\kappa/e)^{\kappa+1}$ for fixed Λ yields $\Lambda^{\kappa+1} / (\kappa + 1)! = \mathcal{O}((e\Lambda/\kappa)^\kappa) = \mathcal{O}(\kappa^{-\kappa})$, which decays faster than any geometric rate. $\square$

B PROOF SKETCH OF PROPOSITION 1

Standard weak-convergence analysis of the Euler–Maruyama scheme for the SDE in Eq. equation 1 gives a local one-step weak error $\mathcal{O}(1/M^2)$ that accumulates with a Grönwall factor: $\epsilon_t^E \leq \frac{K}{M} \exp(Lt)$, where L bounds the Lipschitz constants of the drift/diffusion/jump coefficients and K absorbs the observable’s Lipschitz constant. The restart solver resets the state to the exact analytical conditional mean $\mathbb{E}[S_\tau | C]$ at every integer boundary $\tau = \lfloor t \rfloor$, so error cannot accumulate across intervals: only the elapsed time since the last restart, $t - \lfloor t \rfloor \in [0, 1)$, enters the Grönwall exponent, giving $\epsilon_t^R \leq \frac{K}{M} \exp(L(t - \lfloor t \rfloor)) \leq \frac{K}{M} \exp(L)$, a bound independent of the horizon t . Both schemes retain the $\mathcal{O}(1/M)$ rate; restart improves the constant from $\exp(Lt)$ to $\exp(L)$. $\square$

C ADDITIONAL SPECTRAL-LOSS RESULTS

Table 5 reports the full weight-sensitivity sweep at $T_f = 7$ and the long-horizon comparison ($T_f \in \{7, 14, 21, 28\}$) with and without the spectral loss. At short horizons the point metrics are flat across weights (a gentle prior); the decisive effect is at $T_f = 28$, where the full loss cuts Mean MSE by 46% and raises R^2 from 0.327 to 0.637.

Table 5: **Top:** spectral-loss weight sensitivity at $T_f = 7$ on S&P 500. **Bottom:** long-horizon stability with the full loss ($\lambda = 0.1$) vs. none ($\lambda = 0$). Metrics are Mean MAE/MSE/ R^2 and probabilistic p -MAE/ p - R^2 . **Bold** highlights the decisive long-horizon gain.

Configuration	Mean MAE↓	Mean MSE↓	Mean R^2 ↑	p -MAE↓	p - R^2 ↑
<i>Weight sensitivity ($T_f = 7$):</i>					
$\lambda_{\text{energy}} = 0$ (no reg.)	14.2	930.8	0.953	14.4	0.952
$\lambda_{\text{energy}} = 0.001$	15.3	996.9	0.949	15.1	0.950
$\lambda_{\text{energy}} = 0.01$	14.4	939.6	0.952	14.5	0.952
$\lambda_{\text{energy}} = 0.1$ (default)	14.5	941.2	0.952	14.5	0.952
$\lambda_{\text{energy}} = 1.0$	14.1	944.0	0.952	14.1	0.953
diffusion-only ($\lambda = 0.1$)	14.2	932.0	0.953	14.2	0.953
<i>Long-horizon stability (full $\lambda = 0.1$ vs. none $\lambda = 0$):</i>					
$T_f = 7$ full	14.5	941.2	0.952	14.5	0.952
$T_f = 7$ none	14.2	930.8	0.953	14.4	0.952
$T_f = 14$ full	22.1	2045.6	0.896	20.8	0.902
$T_f = 14$ none	18.4	1503.0	0.924	18.3	0.924
$T_f = 21$ full	24.7	2401.2	0.878	24.1	0.882
$T_f = 21$ none	25.2	2464.4	0.875	24.4	0.880
$T_f = 28$ full	41.8	7161.9	0.637	36.6	0.726
$T_f = 28$ none	54.1	13271.4	0.327	42.0	0.677

Table 6: Runtime (ms) on S&P 500. Neural-MMA avoids the multi-run blow-up of iterative samplers.

Model	Train (ms)	1-run (ms)	10-run (ms)
MLP	65.2	52.2	N/A
GCN	271.3	250.7	N/A
Flow Matching	184.6	275.4	2696.3
Neural-MMA (ours)	183.5	166.8	179.2

D RUNTIME COMPARISON

Table 6 compares wall-clock training and inference cost. Neural-MMA matches the fastest learned baselines for a single run and, crucially, keeps the $K = 10$ multi-run cost nearly identical to a single run (analytical inference adds no NFEs), unlike Flow Matching whose iterative sampling scales poorly.